



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	NANOSCIENCE AND NANOTECHNOLOGY				
Type of Course	ELECTIVE IN MAJOR				
Semester	V				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. A strong grasp of foundational chemistry principles and terminology. 2. Understanding of key physics concepts relevant to materials science.				
Course Summary	This course offers a comprehensive introduction to the interdisciplinary field of nanoscience, covering the fundamental principles, synthesis methods, structural properties, and diverse applications of nanomaterials. Through a blend of theoretical lectures and practical demonstrations, students will gain insight into the unique properties and potential applications of materials at the nanoscale.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the history, scope, definitions, and fundamentals of nanoscience and nanotechnology, including the study of nanomaterials	U	F	Instructor-created exams / Quiz /Assignment
CO2	Apply various methodologies for nanoparticle synthesis and characterization, gaining practical understanding and applications	Ap	C	Class test /Assignment /Quiz
CO3	Grasp the structures and properties of diverse nanomaterials, including carbon, organic, and inorganic	U	P	Assignment/ Class test

	nanomaterials, and understand their applications			
CO4	Examine and apply the principles of photovoltaic energy conversion, targeted drug delivery, and other applications of nanomaterials	An	C	Assignments /Seminar presentation
CO5	Construct ideas about size and shape-dependent catalysis, interactions between biomolecules and nanoparticle surfaces and applications in biology	Ap	M	Assignments /Seminar presentation
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Introduction to Nanoscience		12	20
	1	History and Scope - Feynman's Vision, Moore's Law, Faraday's experiment and Lycopodium cup.	2	
	2	Definitions of Nanoscience and Nanotechnology. Energetics of nanomaterials – kinetic stability, need for surface modification.	2	
	3	Classification of Nanomaterials – 3D, 2D, 1D and 0D, confinement of electrons and phonons, quantum dots.	3	
	4	Surface to volume ratio, Quantum size effect, Surface Effect.	2	
	5	Size-dependent variation in physical- chemical- electronics-catalytic properties.	3	
II	Methods for nanoparticle synthesis and characterization		12	20
	6	Top-down approach - Ball milling, nanolithography.	1	
	7	Bottom-up approach - Growth of nanocrystals in solution – Nucleation and Ostwald ripening. Capping agents – Dispersibility, -OH and -SH based capping agents.	2	
	8	Methods of synthesis - precipitation, sol-gel, hydrothermal, microemulsion, chemical reduction, chemical vapour deposition and self-assembly.	3	

	9	Characterization of nanomaterials by XRD - Theory, factors affecting line broadening, Scherrer equation.	2	
	10	SEM & TEM- Electron wavelength by De-Broglie relation, resolution and resolving power, electron-sample interaction, components, schematic diagram, bright and dark field imaging	2	
	11	AFM- Introduction to scanning probe methods - components, schematic diagram, tapping and scanning modes of AFM)	2	
III	Structure and Properties of Nanomaterials		12	20
	12	Carbon Nanomaterials and the nature of carbon bonds.	1	
	13	Fullerenes - structure of C60, laser ablation synthesis, doping and superconductivity in M3C60.	2	
	14	CNT- chiral, zig-zag and armchair CNT, arc discharge synthesis, electrical and mechanical properties	2	
	15	Graphene- Dependence of edge geometry on electrical, magnetic and optical properties. Oxidative exfoliation synthesis, properties and uses.	2	
	16	Organic nanomaterials – Structure and applications of dendrimers and liposomes. Inorganic nanomaterials	2	
	17	Electrical and optical properties (Eg: TiO ₂ , CdS) Nanomagnetism – Superparamagnetism (Eg: Fe ₃ O ₄). Nanocomposites and their advantages	3	
IV	Applications of nanomaterial		12	20
	18	Principle of photovoltaic energy conversion, TiO ₂ based DSSC- Components and mechanism.	3	
	19	Targeted Drug Delivery using magnetic nanoparticles - functionalization using drug molecules, dispersibility, drug release.	3	
	20	Photodegradation of dyes using TiO ₂ - mechanism	2	
	21	Carbon nanomaterials as adsorbents for remediation and hydrogen storage.	2	
	22	Surface Plasmon Resonance (Eg: Ag or Au nanoparticles) and its application	2	
V	Nanocatalysis and Nanobiology (Open Ended)		12	20
	Open Ended	Size and shape dependent catalysis by nanomaterials, Interaction Between Biomolecules and Nanoparticle Surfaces, Applications of Nanomaterials in Biology.	12	

References:

1. Poole, C. P., & Owens, F. J. *Introduction to nanotechnology*. Wiley-Interscience
2. Vollath, D. *Nanomaterials: An Introduction to Synthesis, Properties and Applications*. John Wiley & Sons.
3. Pradeep, T. *Textbook of nanoscience and nanotechnology*. McGraw-Hill Education.
4. Rao, C. N. R., Müller, A., & Cheetham, A. K. (Eds.). *Nanomaterials chemistry: recent developments and new directions*. Wiley-VCH Verlag GmbH
5. Murty, B., Shankar, P., Raj, B., Rath, B. B., & Murday, J. S. *Textbook of Nanoscience and Nanotechnology*. Springer
6. Kamat, P. V., Murakoshi, K., Wada, Y., & Yanagida, S. *Semiconductor nanoparticles*. In *Handbook of Nanostructured Materials and Nanotechnology* (pp. 291-344). Academic Press.
7. Pathak, Y. V. (Ed.). *Surface modification of nanoparticles for targeted drug delivery*. Cham, Switzerland: Springer
8. Thomas, S., Kalarikkal, N., Oluwafemi, O. S., & Wu, J. (Eds.). *Nanomaterials for solar cell applications*. Elsevier.
9. Varin, R. A., Czujko, T., & Wronski, Z. S. *Nanomaterials for solid state hydrogen storage*. Springer.

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1					1	3		1		1		
CO 2				1	2	3	3		1		3		2
CO 3						1	3		1				
CO 4				2		1	3		1		3		2

CO 5				1	1		3		1		1		3
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Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓